ROGERS STATE UNIVERSITY

Coding and Smart Systems Outreach Program

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Motion Detector with Buzzer

Hardware Components Required:

- Raspberry Pi 4 with GPIO extension
- Breadboard
- One Passive Infrared (PIR) motion sensor
- One Active buzzer
- Wires: red, black, yellow, blue colors

Step-by-step Instructions to Connect Hardware Components:

1. Place the PIR motion sensor near the breadboard.
2. Connect the red wire to GPIO 5V (pin 2).
3. Connect the black wire to GPIO GND (pin 6).
4. Connect the yellow wire to GPIO 17 (pin 11).
5. Plug the active buzzer on the breadboard so that each leg is in a new empty row.
6. Connect the buzzer's longer leg (+) to GPIO 27 (pin 13).
7. Connect the buzzer's shorter leg to the GPIO GND pin (pin 6).
8. Power on Pi.
9. On the screen, **click on the projects folder**.
10. Click on the **motion_sensor.py** file.
11. Write the code.

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Python code (Solution):

```python
# motion_sensor.py

from gpiozero import MotionSensor, Buzzer
from time import sleep

# Define GPIO pins
PIR_PIN = 17
BUZZER_PIN = 27

# Initialize devices
pir = MotionSensor(PIR_PIN)
buzzer = Buzzer(BUZZER_PIN)

print("PIR Motion Detector active. Stabilizing sensor...")
sleep(30)  # Give PIR time to calibrate
print("System ready. Move in front of the sensor to test.")

try:
    while True:
        pir.wait_for_motion()
        print("Motion detected! Beeping 3 times...")

        # Beep 3 times when motion is detected
        for _ in range(3):
            buzzer.on()
            sleep(0.2)   # short beep
            buzzer.off()
            sleep(0.2)   # short pause between beeps

        print("Beeping complete. Waiting for no motion...")

        pir.wait_for_no_motion()
        print("No motion detected. Buzzer OFF.")
        buzzer.off()  # ensure buzzer is off

except KeyboardInterrupt:
    buzzer.off()
    print("\nProgram terminated.")

```